

Hasnain Heickal

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EDUCATION

University of Massachusetts Amherst

MS/PhD in Computer Science and Engineering. Advisor: Dr. Andrew Lan

Massachusetts, USA

Fall, 2021 – Present

University of Dhaka

MS in Computer Science and Engineering

Dhaka, Bangladesh

2012 – 2013

University of Dhaka

B.Sc. (Hons) in Computer Science and Engineering

Dhaka, Bangladesh

2007 – 2011

WORK EXPERIENCE

Summer Intern

Google Inc.

June 2026 - Sept 2026

CA, USA

Worked in BigLake Metastore team under Google Cloud Platform (GCP)

Summer Intern

Educational Testing Service (ETS)

June 2024 - August 2024

NJ, USA

Worked in Automatic Speech and Language Processing (ASLP) team under Product and Innovation Development (PID)

Assistant Professor (On study leave)

February 2018 – Present

Lecturer

Department of Computer Science and Engineering, University of Dhaka

May 2014 – February 2018

Dhaka, Bangladesh

Courses Taught: Advanced Algorithms (MS), Programming Fundamentals, Data Structures, Algorithms, Distributed Systems, Computer Graphics, System Programming, Numerical Analysis, Multi-variable Calculus and Geometry.

Lecturer

Department of Computer Science and Engineering, Eastern University

January 2014 – May 2014

Bangladesh

RESEARCH

Research Interest

Machine Learning for Education, Natural Language Processing

Google Scholar Profile

<https://scholar.google.com/citations?user=oQc7yEAAAAAJ>

Current Research

Automating pedagogical interventions for computer science education

Selected Publications

1. *A Classroom Study of LLM-Generated Feedback Intervention in Introductory Programming* **Hasnain Heickal**, Andrew Lan. IRAISE 2026 in Festival of Learning (AIED, EDM & L@S). ([arxiv](#))
2. *Learning Code-Edit Embeddings to Model Student Debugging Behavior* **Hasnain Heickal**, Andrew Lan. AIED 2025: 91–98. ([arxiv](#))
3. *Generating Feedback-Ladders for Logical Errors in Programming using Large Language Models* **Hasnain Heickal**, Andrew Lan. EDM 2024: 947–951. ([arxiv](#));
4. *An Automatic Abstractive Text Summarization System* Nasid Habib Barna, **Hasnain Heickal**. DUJASE 6(2) (2022): 39–48
5. *End-to-End Speech Synthesis for Bangla with Text Normalization*. Tanzir Islam Pial, Shahreen Salim Aunti, Shabbir Ahmed, **Hasnain Heickal**. CSII 2018: 66–71
6. *Deep Learning Based Bangla Speech-to-Text Conversion*. Md. Tahsin Tausif, Sayontan Chowdhury, Md. Shiplu Hawlader, Md. Hasanuzzaman, **Hasnain Heickal**. CSII 2018: 49–54
7. *Segmentation of Heterogeneous Documents into Homogeneous Components using Morphological Operations*. Nasid Habib Barna, Tisa Islam Erana, Shabbir Ahmed, **Hasnain Heickal**. ICIS 2018: 513–518

8. *A framework for developing graphically programmable low-cost robotics kit for classroom education.* Mehedi Hasan Masum, Tanvir Shahriar Rifat, Saifuddin Md Tareeq, **Hasnain Heickal**. ICETC 2018: 22–26
9. *Computer Vision-Based Real-Time 3D Gesture Recognition Using Depth Image.* **Hasnain Heickal**, Tao Zhang, Md. Hasanuzzaman. International Journal of Image and Graphics 15(1) (2015)

Thesis Supervisions

1. *Automatic Evaluation of Subjective Answers.* Master's thesis of Tisa Islam Erana. Proposed a weighted bipartite matching-based semantic distance measure to group similar answers.
2. *A Hybrid Filtering-based Book Recommender System Using Users' Social Media Activities.* Master's thesis of Shahreen Selim Aunti. Proposed a Hybrid Filtering-based approach that uses users' social media activities to identify their preferences and topics of interest.
3. *Sentiment Analysis using Convolutional Neural Network with Word Embedding and SentiWordNet.* Master's thesis of Tanzir Islam Pial. Proposed a novel approach for sentiment analysis using a CNN that uses word vectors and sentiment information of individual words together.

PROGRAMMING CONTEST EXPERIENCE

Coach of Programming Contest

Department of Computer Science and Engineering, University of Dhaka

2010 – 2021

Department of Computer Science and Engineering, Jahangirnagar University

September 2011 – October 2013

Two teams have qualified to ICPC World Finals under my coaching in 2017 and 2020.

Problem-setting and Judging

2013 – 2021

ICPC Asia West Continent Final Contest

January, 2020

- As Judging Coordinator, coordinated with the judging team to create an original problem set. Team members included judges from Afghanistan, Bangladesh, Iran, India, and Pakistan.

ICPC Dhaka Regional

2013, 2017 – 2019

- As Judging Director on 2019, managed the whole judging team consisting 23 persons.
- As Problem-setter and Judge, created few original problems, wrote solutions, tests etc.

National Collegiate Programming Contest (NCPC)

2014 - 2015, 2017 - 2020

- As Judging Coordinator on 2018 - 2020, coordinated with the judging team to create an original problem set.
- As Problem-setter and Judge, created few original problems, wrote solutions, tests etc.

Team Leader, Bangladesh Informatics Olympiad Team

2014

International Olympiad of Informatics (IOI)

Taipei, Taiwan

Competitive Programming Achievements

- 1st position in IUPC by BGC Trust University, Bangladesh in 2012. Team - DU Angry Birds
- 1st position in NCPC by University of Dhaka, Bangladesh in 2011. Team - DU Resonance
- 1st position in IUPC by IUT, Bangladesh in 2011. Team - DU Resonance
- 5th position in ICPC Dhaka Regional by NSU, Bangladesh in 2011. Team - DU Resonance
- 4th position in ICPC Dhaka Regional by NSU, Bangladesh in 2010. Team - DU Resonance
- 6th Position in ICPC Amritapuri Regional by Amrita Vishwa Vidyapeetham, India in 2010. Team - DU Resonance
- 2nd position in NCPC by SUST, Bangladesh in 2010. Team - DU Resonance
- 4th position in ICPC Amritapuri Regional by Amrita Vishwa Vidyapeetham, India in 2009. Team - DU Dark Shadows

Online Problem Solving

- Solved more than 700 problems in Valladolid Online Judge. Profile: [hhjami](#)
- Maximum rating in Topcoder – 1692 (Yellow). Profile: [Jami.CSEDU](#)
- Maximum rating in Codeforces – 2128 (International Master). Profile: [Jami.CSEDU](#)

SKILLS

Programming Languages: C/C++, Python, Java, PHP, JavaScript, SQL, QBasic, Assembly, Lisp, Prolog.

Key Concepts: Data Structures, Algorithms, NLP, Machine Learning, Databases, Object Oriented Programming

Tools: Vim, Git, Google Cloud Platform, VS Code, Eclipse, Netbeans

Libraries: PyTorch, OpenGL, Weka, OpenNI, TensorFlow

Frameworks: React, Node.js, WordPress, Laravel

Operating Systems: Ubuntu, MacOS, Windows, Fedora, CentOS

EXTRA CURRICULAR ACTIVITIES

- Champion in Regional Mathematics Olympiad, Chittagong, 2006.
- Second Runner Up in National Mathematics Olympiad, Bangladesh, 2006.
- Runner Up in Inter School Debate Championship, 2002, organized by Drishty, Chittagong.
- Can play several musical instruments such as Violin, Guitar, Mouth Organ, etc.